Lesson plan · for the teacher

Algorithms: exact instructions

Programming · Level 1 · Lesson 1

Time
40 min

Materials
Chalk for a grid on the floor or the board; Paper arrow cards: forward, turn left, turn right; Your phone (optional) with this lesson downloaded

Learning goals

- Say what an algorithm is.
- Explain why a computer needs exact instructions.
- Play the human robot and write its program.

Activity: The human robot

1. Warm-up (5 min): Ask a volunteer to draw on the board only what the class says. Ban the word “square”. Laugh at what appears, then agree on exact steps.
2. Teach (5 min): Define an algorithm: exact steps, in order, that finish. Show the three cards: forward one square, turn left, turn right (turning on the spot).
3. Play (15 min): Draw a 5 × 5 chalk grid with a start and a flag. Pairs lay out cards; a robot pupil follows them exactly while the class watches.
4. Debug (10 min): When the robot misses, stop at the first wrong move. The pair finds the card, changes only that one, and runs again.
5. Wrap-up (5 min): Ask why the robot never “fixes” a mistake by itself. Then give the exit ticket.

No device is needed. If you have one phone, show the robot app at the end as “the same game, on a screen”.

Discussion

- Which instruction caused the most trouble, and why?
- Where do pupils already follow algorithms in other subjects?
- Why can’t a computer guess what we meant?

Exit ticket

- Write an exact algorithm for one thing you did this morning.

Try it in class tomorrow
Play the human robot: one pupil follows only the class’s exact instructions to walk to the door. Each time the robot does something silly, ask: which instruction wasn’t exact?

Worksheet · for pupils

Algorithms: exact instructions

Programming · Level 1 · Lesson 1

Name _____ Class _____ Date _____

1 **The robot faces up. The flag is one square up and one square to the right. Which program works?**

◯ Turn right, forward, forward ◯ Forward, forward, turn right ◯ Forward, turn right, forward

2 **Put this algorithm for washing your hands in order.**

☐ Put soap on them

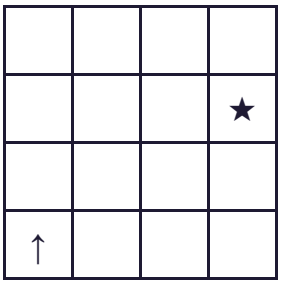
☐ Rub them together for 20 seconds

☐ Dry your hands

☐ Wet your hands

☐ Rinse off the soap

3 **Build a program that walks the robot to the flag. Tap a card in your program to repeat it, up to 4 times.**



Cut out cards (↑ forward, ↶ turn left, ↷ turn right) and lay out a program that takes the robot to the ★. Write it here:

_____

_____

4 **If an instruction is a little unclear, a computer guesses what you meant.**

◯ True ◯ False

5 **Which of these are algorithms?**

☐ The steps of long division ☐ “Have a nice day” ☐ Directions from the bus stop to the school

☐ A recipe for bread

Answers · for the teacher

Algorithms: exact instructions

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1 **Forward, turn right, forward**

Up one square first, then turn to face right, then one more square.

2 **Top to bottom: 2, 3, 5, 1, 4**

Each step needs the one before it: you can't rinse off soap you haven't put on.

3 **Any program that takes the robot to the ★ with 4 cards or fewer.**

Forward twice, turn right, then forward three times: three cards.

4 **False**

A computer never guesses: it does exactly what the program says.

5 **A recipe for bread; Directions from the bus stop to the school; The steps of long division**

A recipe, directions and long division are exact steps, in order, that finish with a result.

✎ For you to think about

Which instruction do you give pupils every day that could be more exact? Rewrite it as numbered steps.